

Device: cuda

Torch dtype: torch.float32

Bundle sizes: (48, 48, 48)

Coarse load map C:

Let the hypercube bit directions be partitioned into three bundles

I_x, I_y, I_z .

For a microscopic local cell a , define its coarse address

$\pi(a) = (\text{number of active directions in } I_x,$
 number of active directions in $I_y,$
 number of active directions in $I_z).$

For a microscopic Hodge-Dirac load density $\rho_D(a)$,

$(C \rho_D)_{\ell} = (1/Z_{\ell}) \sum_{a : \pi(a)=\ell} \rho_D(a).$

The null space $\ker(C)$ consists of microscopic load patterns that are invisible to the macroscopic load observable. This is the operational definition of the gauge/vacuum redundancy $G = \ker(C)$.

Load quotient leaves: 117,649

Source coordinate: $(\text{np.int64}(24), \text{np.int64}(24), \text{np.int64}(24))$

Source index: 58824

Undirected quotient edges: 345,744

/tmp/ipykernel_1104/2474588782.py:175: UserWarning: Sparse invariant checks are implicitly disabled. Memory errors (e.g. SEGFAULT) will occur when operating on a sparse tensor which violates the invariants, but checks incur performance overhead. To silence this warning, explicitly opt in or out. See

``torch.sparse.check_sparse_tensor_invariants.__doc__`` for guidance. (Triggered internally at /pytorch/aten/src/ATen/Context.cpp:760.)

```
L = torch.sparse_coo_tensor(indices, values, (N, N), device=device).coalesce()
```

CG iter 1 | relative residual 4.082e-01

CG iter 100 | relative residual 1.888e-05

CG converged at iter 119 | relative residual 9.317e-07

CG elapsed: 0.05 s

--- Green kernel fit ---

$K_D(r) \approx A/r + B$

A = 7.85345551e-02

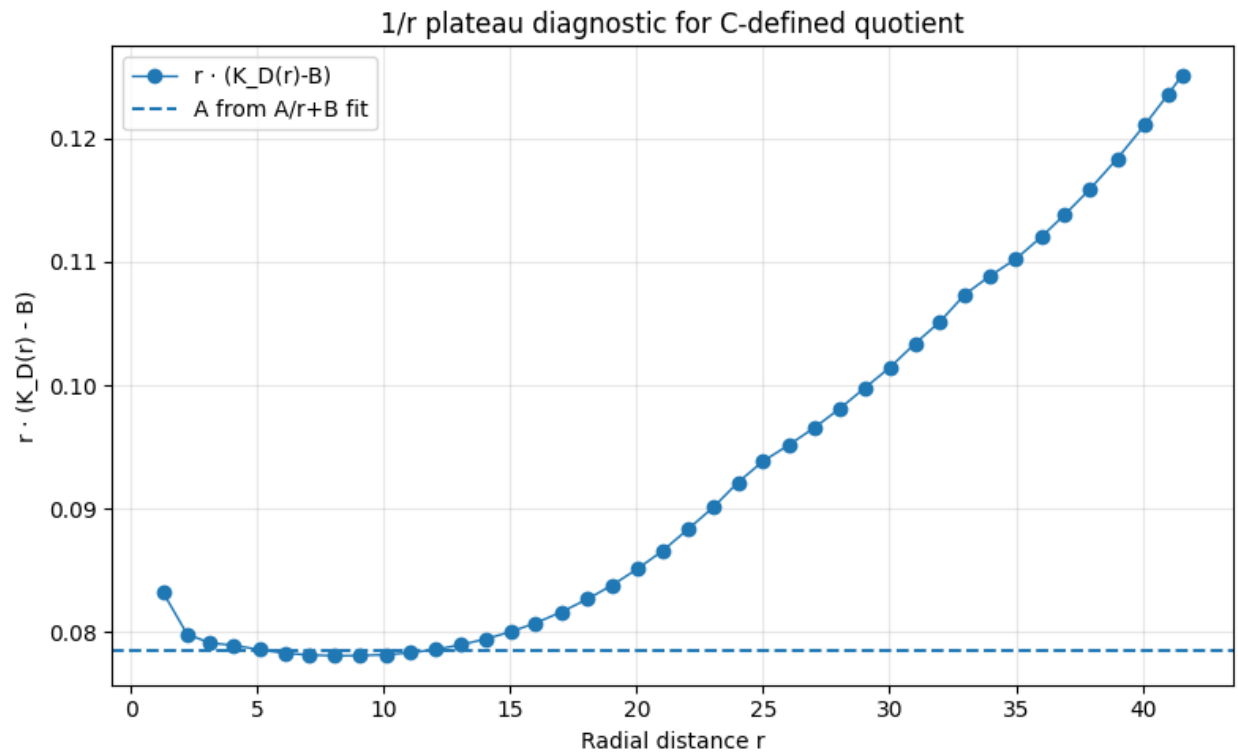
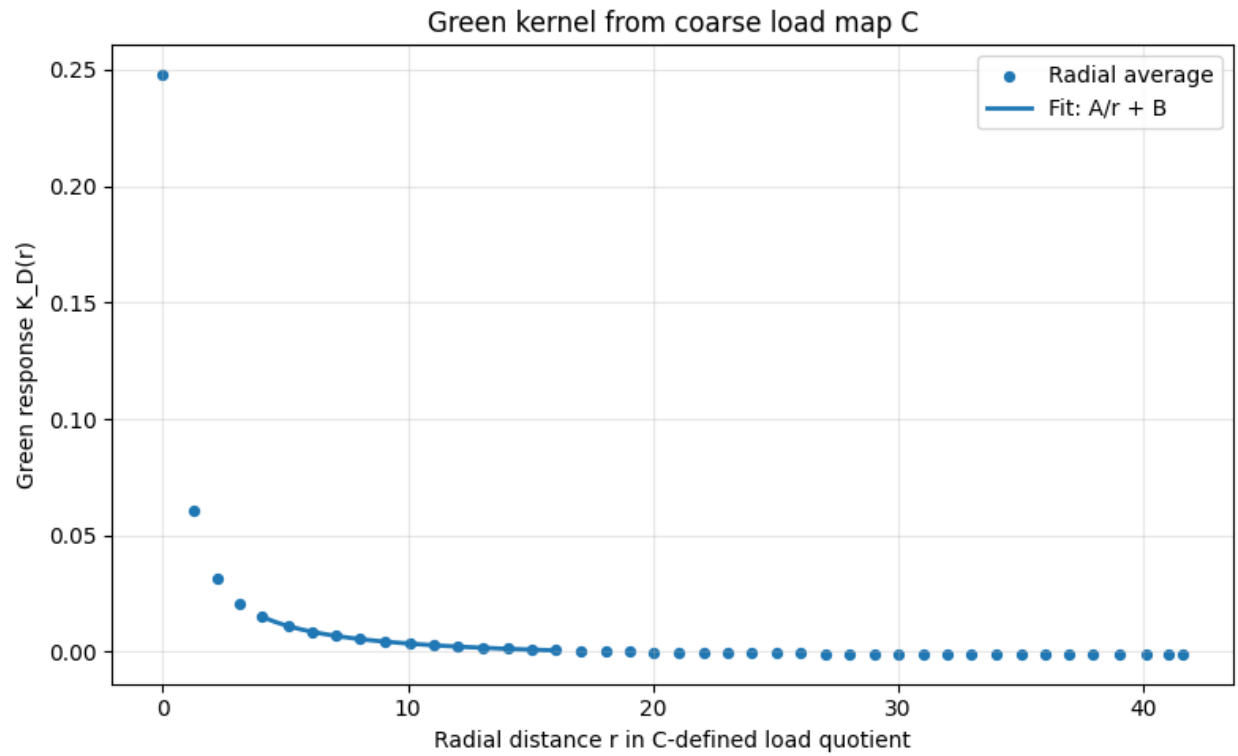
B = -4.31376118e-03

$1/(4\pi) = 7.95774715e-02$

Relative error in A = 1.311%

$R^2 = 0.99982638$

Fit window: r in [4, 15.84]



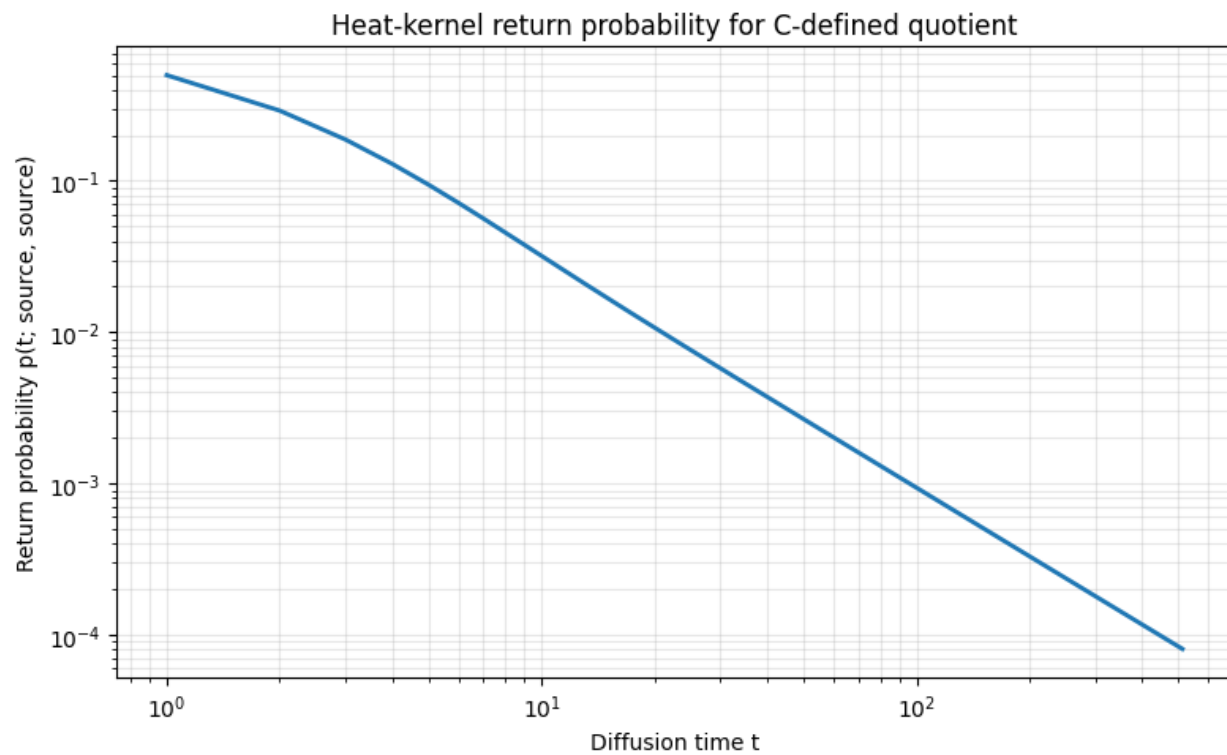
Heat diffusion elapsed: 0.03 s

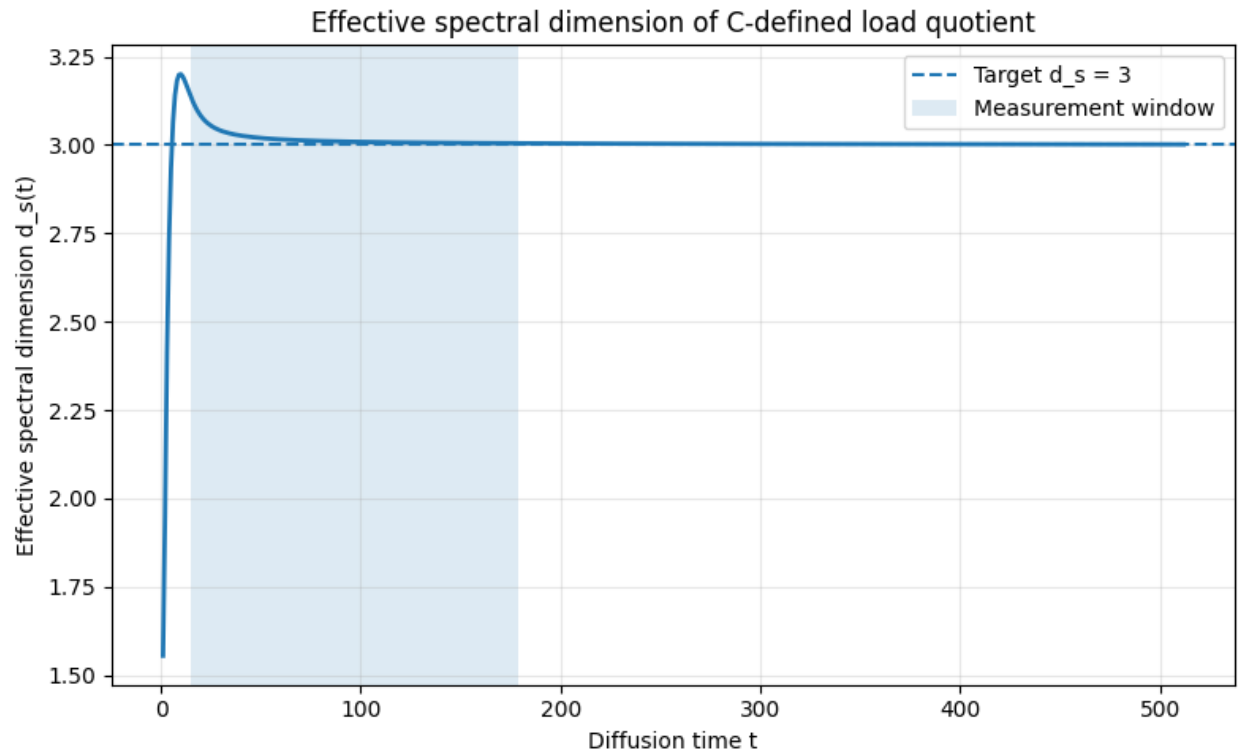
--- Heat-kernel spectral dimension ---

Window: t in $[15, 179]$

Median $d_s \approx 3.0088$

Mean $d_s \approx 3.0171$





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SUMMARY

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Green $1/r$ fit R^2 : 0.99982638 -> PASS

A relative error vs $1/(4\pi)$: 1.311%

Median spectral dimension: 3.0088 -> PASS

Output directory:

/content/admissible_load_C_results

Files written:

- cg_convergence.csv
- coarse_load_map_C_definition.csv

- green_kernel_fit_summary.csv
- green_kernel_plateau_diagnostic.png
- green_kernel_radial_average.csv
- green_kernel_radial_fit.png
- heat_kernel_spectral_dimension.csv
- heat_return_probability.png
- spectral_dimension_estimate.png
- summary.csv

Manuscript-safe interpretation:

This experiment defines an explicit coarse load map C by grouping hypercube bit directions into three balanced macroscopic load bundles and quotienting out microscopic binomial degeneracy through capacity normalization. The null space $G=\ker(C)$ is the microscopic sector invisible to macroscopic load propagation.

The resulting load quotient is tested by computing $K_D=L_{\text{grav}}^+$. A simultaneous $A/r+B$ Green-kernel scaling window and heat-kernel spectral dimension $d_s\approx 3$ support the claim that this C -defined load sector has a Newtonian weak-field continuum limit.

This still does not prove that least admissible structural action uniquely selects this C . The remaining theorem is to derive this coarse load map, or its three-dimensional spectral equivalent, from the hypercube admissibility and least-action constraints rather than imposing the three-bundle decomposition.